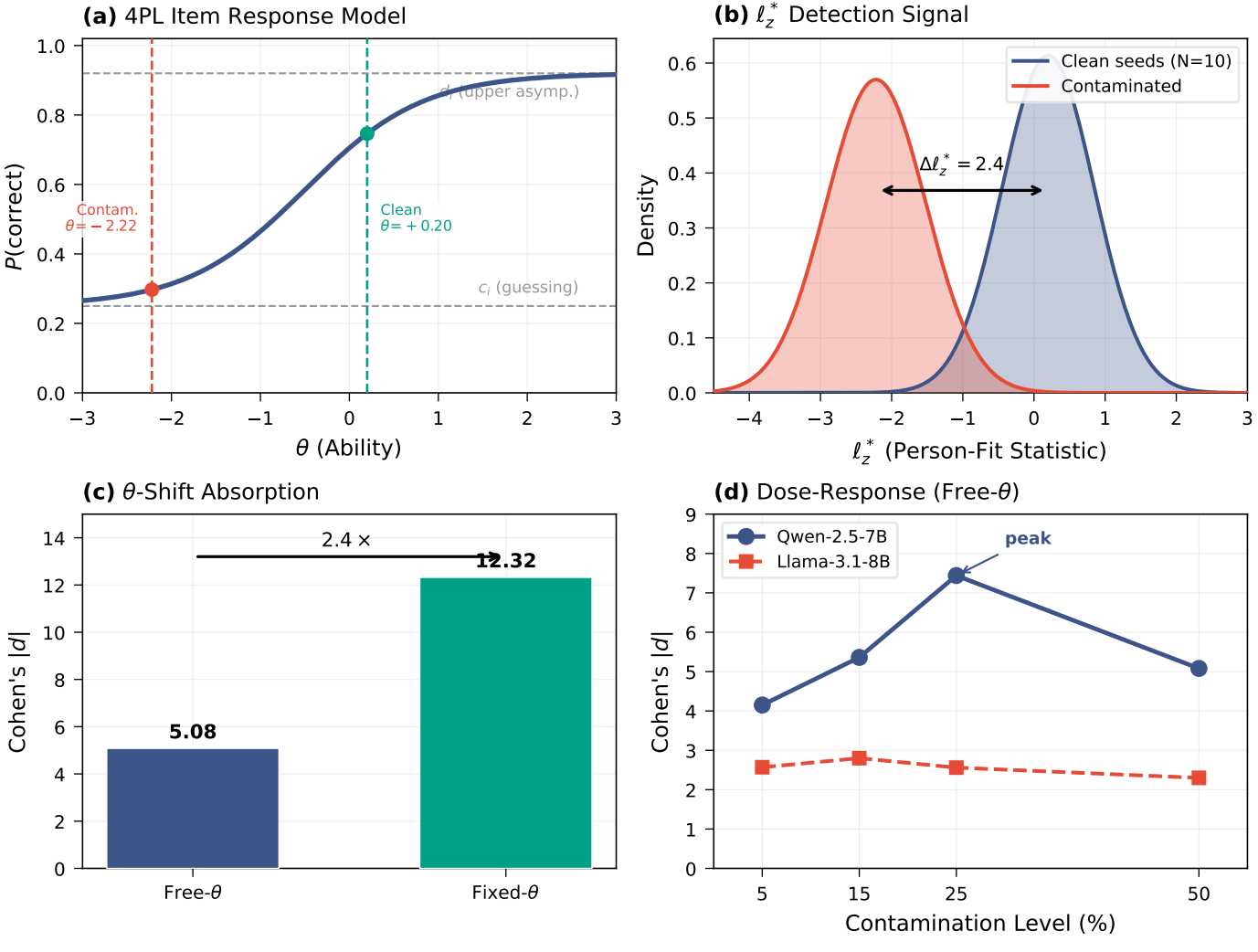
(a) 4PL Item Response Model
P(correct)
θ (Ability)
Contam.
θ = − 2.22
Clean
θ = + 0.20
d_i (upper asymp.)
c_i (guessing)
(b) ℓ_z^* Detection Signal
Density
ℓ_z^* (Person-Fit Statistic)
Clean seeds (N=10)
Contaminated
$\Delta\ell_z^* = 2.4$
(c) θ-Shift Absorption
Cohen's |d|
2.4 ×
12.32
5.08
Free-θ
Fixed-θ
(d) Dose-Response (Free-θ)
Cohen's |d|
Contamination Level (%)
Qwen-2.5-7B
Llama-3.1-8B
peak